L-P-50b
November 25, 1966

SUPERSEDING
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November 2, 1965

FEDERAL SPECIFICATION

# PADS, SCOURING

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers the requirements for scouring pads (see 6.1).

**1.2 Classification.**

1.2.1 *Types, sizes, and classes.* Scouring pads covered by this specification shall be of the following types, sizes, and classes, as specified (see 6.2):

Type I—Plastic (ball type).
  Size 1—4 by 3-1/2 inches.
Type II—Abrasive pad (nonwoven mat).
  Class 1—Light duty.
    Size 1—6 by 9-1/2 by 3/8 inch.
  Class 2—Heavy duty.
    Size 1—6 by 9-1/2 by 3/8 inch.
Type III—Plastic sponge (with abrasive back).
  Size 1—3 by 4-1/2 by 1/2 inch.
  Size 2—3-1/2 by 5 by 1-1/2 inches.
Type IV—Cloth, woven (abrasive coated).
  Size 1—5-1/2 by 5-3/4 inches.
Type V—Cellulose sponge (with nonwoven abrasive back).
  Size 1—3-3/4 by 6-3/4 by 1 inch.

## 2. APPLICABLE DOCUMENTS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitations for bids or request for proposal, form a part of this specification to the extent specified herein:

*Federal Specifications:*

O-S-598—Sodium Hydroxide, Technical.
O-S-642—Sodium Phosphate, Tribasic, Technical; Anhydrous, Dodecahydrate and Monohydrate.
P-C-435—Cleaning compound, General Purpose (Powdered, Heavy Duty).
P-C-442—Cleaning Compound, Soap Abrasive, Paste, for Painted Surfaces.
P-D-220—Detergent, General Purpose.
P-D-223—Detergent, General Purpose, Liquid, Heavy Duty (Built, Synthetic).
P-D-225—Detergent, General Purpose, Solvent Type.
TT-T-291—Thinner; Paint, Volatile Spirits (Petroleum-Spirits).
CCC-T-191—Textile Test Methods.
PPP-B-566—Boxes, Folding, Paperboard.
PPP-B-636—Box, Fiberboard.
PPP-B-665—Boxes; Paperboard, Metal Stayed (Including Stay Material).
PPP-B-676—Boxes, Setup.

*Federal Standards:*

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Test Method Std. No. 406—Plastics: Methods of Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside



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the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publications.** The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

*U.S. Department of Commerce Commercial Standard (CS):*

CS217-59—Grading of Abrasive Grain on Coated Abrasive Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.)

## 3. REQUIREMENTS

**3.1 General requirements.** The materials used in the manufacture of scouring pads shall conform to the applicable requirements specified herein. Since scouring pads are generally used for cleaning cooking utensils and other containers used for food supplies, any components, chemicals, finishing materials, dyes, and/or pigments used in the manufacture of the pads shall show good fastness to cleaning solutions and shall not contain any toxic ingredient that may be extracted in the cleaning solutions under normal use. The scouring pads shall be free from harmful bacteriological effects and shall not contain any ingredient which may cause dermatitis or other harmful effects to personnel using the pad. Scouring pads will be accepted on the basis of the contractor's certification of compliance to the above requirements and the requirements specified in applicable paragraphs of this specification.

**3.2 Materials.**

3.2.1 *Type I.*

3.2.1.1 *Polypropylene.* The material from which the monofilaments for type I pads are extruded, shall be unstabilized polypropylene resin meeting the physical properties specified in table I, when tested as specified in 4.2.3. The monofilaments shall be flat, 0.036 ± 0.003 inch wide and not less than 0.009 inch in thickness.

3.2.1.1.1 *Weight.* The finished pad shall weigh no less than 21.6 grams (g.).

TABLE I. *Physical properties for type I scouring pads*

| Characteristics | Unit of measurement | Value required |
|---|---|---|
| Tensile strength | Pounds per square inch | 6,000 to 10,000 |
| Stress cracking | Flexures | 800,000, minimum |
| Softening point | Degrees Fahrenheit | 280, minimum |
| Melting point | Degrees Fahrenheit | 310, minimum |

3.2.2 *Type II, classes 1 and 2.*

3.2.2.1 *Mineral.* The aluminum-oxide abrasive shall be of good quality artificial corundum (purified corundum), free from extraneous material. The grade of the abrasive shall be very fine.

3.2.2.1.1 *Grading control.* The grade of the aluminum oxide shall be controlled by sedimentation methods (see 4.3.9). Very fine grade shall average 34 ± 5 microns. Small additions, 10 to 15 percent of total weight of abrasive, of titanium oxide (coloring pigment) smaller than 34 microns will be permitted.

3.2.2.2 *Filament.* The filament shall be of good quality crimped, cut nylon (type ny-



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lon 66 or 6) having a nominal staple length of 1 to 8 inches.

3.2.2.2.1 *Class* 1. For class 1 scouring pads, the filaments shall be 15 ± 5 denier.

3.2.2.2.2 *Class* 2. For class 2 scouring pads the filaments shall be 58 ± 10 denier.

3.2.2.3 *Adhesive.* The adhesive used as a binder shall be a water insoluble nonthermoplastic resin. Small amounts of modifying additives, coloring dyes, or pigments are acceptable. The adhesive shall show negligible attack to chemicals, cleaning solutions, cold and boiling water when tested as specified in 4.3.7.

3.2.2.4 *Thickness.* The thickness for class 1 scouring pads shall be not less than 0.250 inch, and the thickness for class 2 scouring pads shall be not less than 0.350 inch when measured as specified in 4.3.1.

3.2.2.5 *Weight.* The weight of the material for classes 1 and 2 scouring pads shall be within the normal commercial tolerances of the manufacturer. The normal material is a light weight high loft mat.

3.2.3 *Type III.*

3.2.3.1 *Polyesterurethane foam.* The sponge portion shall be made of uncompressed polyesterurethane foam meeting the physical properties specified in table II. The foam shall be of fine pore texture and shall have pores, in general, ranging from 0.018 to 0.028 inch in diameter. The foam shall show no appreciable change in texture or volume when wet, and shall be resistant to deterioration by soap or detergent, cleaning solutions, bleach, grease, bacterial attack, or other common causes. Testing shall be as specified in 4.2.3.

TABLE II. *Physical properties for type III scouring pads*

| Property | Value required |
|---|---|
| (Urethane foam component) | |
| Density, lbs. per cu. ft. | 1.90 to 2.20 |
| Rebound resilience (ball drop) percent, minimum. | 25 to 35 |
| Ash maximum percent by weight. | 0.5 |
| Tensile strength, p.s.i. minimum: Length direction. | 20 |
| Water absorption minimum percent by weight: | *Size* 1 / *Size* 2 |
| Initial weight of $H_2O$ absorbed. | 2840 / 2950 |
| Absorption after blotting. | 760 / 860 |
| Tensile strength (combined foam and abrasive coating) lbs. p.s.i minimum.[1] Length direction. | 25 / 30 |

[1] Foam shall be moist containing approximately 50 percent water.

3.2.3.2 *Mineral.* The aluminum oxide abrasive shall be of good quality artificial corundum (purified corundum), free from extraneous material. The size of aluminum oxide grain shall be 60 to 80 mesh with a hardness of 9 Moh's (approximately 2,000 knoop).

3.2.3.2.1 *Grading control.* The grading control and tolerances of the abrasive grain shall be as specified in 4.3.9.

3.2.3.3 *Adhesive.* The adhesive material shall be of a good quality waterproof resin or rubber resin combination suitably formulated to retain its holding power to both the abrasive grit and foam, and to withstand the various service conditions that will be encountered in its use as a scouring pad. The adhesive shall show negligible attack to chemicals, cleaning solutions, cold, and boiling water when tested as specified in 4.3.7.

3.2.3.4 *Thickness.*

3.2.3.4.1 *Size* 1. Size 1 scouring pads shall have an overall thickness of 1/2 ± 1/32 inch. The foam portion shall range 0.375 to 0.437 inch in thickness, and the abrasive coating shall range 0.0625 to 0.125 inch in thickness.

3.2.3.4.2 *Size* 2. Size 2 scouring pads shall have an overall thickness of 1-1/2 ± 1/16 inch. The foam portion shall range 1.375 to 1.437 inches in thickness, and the abrasive coating shall range 0.0625 to 0.125 inch in thickness.

3.2.4 *Type IV.*

3.2.4.1 *Base cloth.* The base cloth shall



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TABLE III. *Physical requirements for terry cloth*

| Tear strength | Texture (yarns per inch) | | | | Loops | Weight, oz./sq. yd. | |
|---|---|---|---|---|---|---|---|
| Pendulum method | Total warp | Ground warp | Pile warp | Filling | Per sq./in. | Uncoated | Coated |
| Warp | | | | | | | |
| Grams (min.) | (*Min.*) | (*Min.*) | (*Min.*) | (*Min.*) | (*Min.*) | (*Min.*) | (*Min.*) |
| 1600 | 70 | 35 | 35 | 32 | 370 | 9.6 | 15.8 |

consist of filling, ground warp, and pile warp single ply cotton yarns, woven with a terry weave meeting the physical requirements specified in table III. Testing shall be as specified in 4.2.3.

3.2.4.2 *Coating compound.* The coating compound shall be a mixture of silica sand abrasive and water insoluble thermosetting plastic resin suitably formulated for the purpose intended. The solubility of the coating compound to water shall not exceed 4 percent. Testing shall be as specified in 4.2.3.

3.2.5 *Type V.*

3.2.5.1 *Cellulose sponge.* The sponge portion of the pad shall be made of a regenerated cellulose with suitable compounding ingredients meeting the physical properties specified in table IV, when tested as specified in 4.2.3. The sponge shall be uncompressed, of coarse pore texture, and shall have pores, in general, ranging from 3/32 to 3/4 inch in diameter.

3.2.5.2 *Abrasive mat.* The abrasive portion of the pad shall be a nonwoven mat made from the materials specified in 3.2.2.1, 3.2.2.2, 3.2.2.2.1, and 3.2.2.3.

TABLE IV. *Physical properties for type V pads*

| Property | Value required |
|---|---|
| (Cellulose sponge component). | |
| Tensile strength, original, minimum, lbs. p.s.i. | |
| Length direction. | 25 |
| Water absorption, minimum percent by weight: | |
| Initial weight of $H_2O$ absorbed. | 1500 |
| Absorption after blotting. | 300 |
| Ash, maximum percent by weight. | 5.0 |
| pH range. | 6.0 to 9.0 |
| (Combined sponge and nylon abrasive mat). | |
| Adhesion strength minimum lbs. (2 inch width). | |
| Between sponge layer and abrasive mat. | 4 |
| Tensile strength, lbs. p.s.i. (sponge shall be moist containing approximately 50 percent water). | |
| Length direction. | 25 |
| Elongation, percent. | 20 to 28 |

3.2.5.3 *Thickness.* Type V scouring pads shall have an overall thickness of 1 ± 1/16 inch. The cellulose sponge portion shall range 0.700 to 0.800 inch in thickness, and the nonwoven mat shall range 0.250 to 0.350 inch in thickness.

3.3 **Construction.**

3.3.1 *Type I.* The plastic monofilaments used in the construction of type I scouring pads shall be knitted together into a seamless stocking approximately 32 inches in length and 6 inches in width, using a 1/4 inch loop type stitch. The filament stocking shall be securely tied at one end with twine and folded up into a ball. The folded portion shall be turned in and the remaining end of the filament stocking securely tied with twine.

3.3.2 *Type II.* Classes 1 and 2 scouring pads shall be constructed in such a manner that the mineral (3.2.2.1), and the adhesive (3.2.2.3), materials are evenly distributed on both sides and through the entire depth of the pads so as to obtain essentially the same abrasive characteristics when used on either side.

3.3.2.1 *Matting.* The finished scouring



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pads shall not mat or wad easily while in use.

3.3.3 *Type III.* Sizes 1 and 2 scouring pads shall be manufactured in accordance with standard commercial practice. The mixture of mineral (3.2.3.2), and adhesive (3.2.3.3), materials shall be suitably formulated and evenly applied to the surface on one side of the foam sponge. The ply strength of the abrasive coating shall be greater than the internal strength of the foam sponge.

3.3.4 *Type IV.* Type IV scouring pads shall be constructed by uniformly coating the base cloth specified in 3.2.4.1, on both sides with approximately equal amounts of the coating compound specified in 3.2.4.2.

3.3.5 *Type V.* Type V scouring pads shall be constructed by laminating the nonwoven abrasive mat (3.2.5.2), to the surface on one side of the cellulose sponge without the use of adhesive. The ply strength of the nonwoven abrasive cellulose sponge laminate shall be greater than the internal strength of the sponge layer.

3.4 **Sizes and measurements.** The sizes and measurements of the finished scouring pads shall be in accordance with table V. The dimensions specified in table V, for types III and V pads shall be determined as described in 4.3.2.

3.5 **Water, chemical, and alkali resistance.** Any sample types II, III, IV, and V pads showing more than negligible softness, tackiness, or loss of abrasive and adhesive substance, when tested as specified in 4.3.7 and 4.3.8, shall be cause for rejection of the lot represented.

3.6 **Identification marking.** The scouring pads shall be marked with the name of the manufacturer, or an abbreviation thereof, or trademark of such known character that the source of manufacture can be readily identified.

3.7 **Workmanship.** The scouring pads shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the acceptable quality levels (AQL's).

## 4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling and inspection.** Sampling and inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the

TABLE V. *Dimensions of scouring pads*

| Type | Length | Tolerance | Width | Tolerance | Thickness | Tolerance |
|---|---|---|---|---|---|---|
| | *(Inches)* | *(Inch)* | *(Inches)* | *(Inch)* | *(Inches)* | *(Inch)* |
| I | 3-1/2 | − 1/2 | 4 | − 1/2 | — | — |
| II, class 1 | 9-1/2 | ± 1/8 | 6 | + 1/8 − 1/16 | 3/8 | [1] |
| II, class 2 | 9-1/2 | ± 1/8 | 6 | + 1/8 − 1/16 | 3/8 | [1] |
| III, size 1 | 4-1/2 | + 1/8 − 1/16 | 8 | + 1/8 − 1/16 | 1/2 | [2] |
| III, size 2 | 5 | ± 1/16 | 3-1/2 | ± 1/16 | 1-1/2 | [2] |
| IV | 5-3/4 | ± 1/16 | 5-1/2 | ± 1/16 | — | — |
| V | 6-3/4 | ± 1/16 | 3-3/4 | ± 1/16 | 1 | [3] |

[1] See 3.2.2.4.
[2] See 3.2.3.4.
[3] See 3.2.5.3.



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same type, size, and class submitted for inspection and delivery at one time.

4.2.1 *Inspection of components and materials.* In accordance with 4.1, the supplier is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 *Examination of the end item.* Examination of the end item shall be in accordance with the classification of defects, inspection levels, and AQL's set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of scouring pads of the specified type, class, and size for examination under 4.2.2.1 and 4.2.2.2; in units of intermediate packages for examination under 4.2.2.3; and in units of shipping containers under 4.2.2.4.

4.2.2.1 *Examination of the end item for defects in material, construction, appearance, and workmanship.* The sample unit for this examination shall be one scouring pad.

| Examine | Defect |
|---|---|
| Type I: | |
| Material | Not fabricated of the specified material. |
| | Not of continuous filament, defect affecting serviceability. |
| Design and construction | Not as specified, ends of filament not tied securely. |
| Types II, III, and V: | |
| Materials | Not as specified. |
| Appearance | Wrinkles in material. |
| | Excessively large, hard particles of adhesive and abrasive. |
| | Excessively large, thin areas or gouged surfaces. |
| | Uncoated areas. |
| | Holes, breaks, creases, or tears affecting serviceability. |
| | Uneven or ragged edges. |
| | Any disagreeable odor. |
| Component parts (where applicable) | Missing, not as specified. |
| | Not soft and pliable. |
| Type IV: | |
| Materials | Not as specified. |
| Appearance | Cut, hole, tear affecting serviceability. |
| | Terry loops missing, broken, or missing yarns. |
| | Uncoated areas. |
| | Uneven or ragged edges. |
| | Excessive hard particles of adhesive and abrasive. |
| | Any disagreeable odor. |

4.2.2.2 *Examination of the end item for dimensional defects.* The sample unit for this examination shall be one scouring pad.

| Examine | Defect |
|---|---|
| Dimensions[1] | Length, width, diameter, or thickness (as applicable), less than the dimensions specified in table V. |

[1] Types III and V scouring pads shall be immersed in water as specified in 4.3.2, prior to the dimensional examination.

4.2.2.3 *Examination of the end item for defects in count.* The sample unit for this examination shall be one unit (intermediate) package or container, as applicable. The count shall be not less than the specified or indicated quantity.

4.2.2.4 *Examination of preparation for delivery.* An examination shall be made to determine that packaging, packing, and marking comply with the requirements of section 5. The sample unit for this examination shall be one shipping container fully prepared for delivery.

| Examine | Defect |
|---|---|
| Packaging (as applicable) | Not in accordance with requirements. |
| | Not unit packaged as specified. |
| | Mixed type, class, or size of scouring pads in same unit container. |
| | Packaging material not as specified; closures not accomplished by specified or required methods or materials. |



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| Examine | Defect |
|---|---|
| Packing (as applicable) (cont'd) | Not in accordance with requirements. |
| | Arrangement or number of unit packages per container not in accordance with requirements. |
| | Container material not as specified; closures not accomplished by specified or required methods or materials. |
| | Container material broken, dented, or otherwise damaged to the extent that serviceability will be affected. |
| Markings | Interior or exterior markings (as applicable), illegible, incorrect, incomplete, omitted, or not in accordance with requirements. |

4.2.2.5 *Inspection levels and AQL's for examinations.* The inspection levels, for determining the sample size and the AQL's, expressed in defects per 100 units, shall be as follows:

| Inspection paragraph | Inspection levels | AQL's |
|---|---|---|
| 4.2.2.1 | S-4 | 4.0 |
| 4.2.2.2 | S-2 | 4.0 |
| 4.2.2.3 | S-2 | 4.0 |
| 4.2.2.4 | S-2 | 2.5 |

4.2.3 *Testing of the end item.* The end item shall be tested for the applicable characteristics as indicated in table VI, herein, from each lot presented for examination for each type, class, and size of scouring pad. The sample unit as referenced herein, is defined as the amount of material required to perform all tests one time. The sample unit, randomly selected, shall consist of 10 scouring pads. Each sample unit shall be drawn from a different container for better representation of the lot. Three sample units, randomly selected, shall be tested from each lot with no evidence of failure to meet unit or average requirements, as applicable. The lot size shall be expressed in units of scouring pads of the specified type, class, and size.

TABLE VI. *Instructions for testing*

| Characteristics | Specification reference | | | | | | No. determinations per unit | Results reported as | |
|---|---|---|---|---|---|---|---|---|---|
| | Requirement reference | Test paragraph | Test method | | Requirements applicable to | | | Numerically to nearest | Pass or fail |
| | | | CCC-T-191 | Fed. Std. No. 406 | Individual unit | Lot average | | | |
| (Type I) | | | | | | | | | |
| Tensile strength | Table I | — | — | 1013 | — | X | 1 | 0.1 lb. | — |
| Stress cracking | Table I | — | — | 1031 | — | X | 1 | — | X |
| Heat-distortion temperature | Table I | — | — | 2011 | — | X | 1 | — | X |
| (Type II) | | | | | | | | | |
| Filament denier[1] | 3.2.2.2.1; 3.2.2.2.2 | — | — | — | — | — | — | — | — |
| Thickness | 3.2.2.4 | 4.3.1 | — | — | — | X | 1 | — | X |
| Resistance to water, chemicals, and cleaning solutions | 3.2.2.3; 3.5 | 4.3.7; 4.3.8 | — | — | X | — | 1 | — | X |
| (Type III) | | | | | | | | | |
| Tensile strength, (foam component) | Table II | — | 5100[2] | — | — | X | 1 | 0.1 lb | — |
| Ash, (foam component) | Table II | 4.3.5 | — | — | — | X | 1 | 1.0 percent | — |
| Water absorption, (foam component) | Table II | 4.3.3 | — | — | — | X | 1 | 1.0 percent | — |



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TABLE VI. *Instructions for testing (cont'd)*

| Characteristics | Specification reference | | | | | | No. determinations per unit | Results reported as | |
|---|---|---|---|---|---|---|---|---|---|
| | Requirement reference | Test paragraph | Test method | | Requirements applicable to | | | | |
| | | | CCC-T-191 | Fed. Std. No. 406 | Individual unit | Lot average | | Numerically to nearest | Pass or fail |
| Absorption after blotting, (foam component) | Table II | 4.3.4 | — | — | — | X | 1 | 1.0 percent | — |
| Tensile strength, (combined foam and abrasive coating) | Table II | — | 5100[2] | — | — | X | 1 | 0.1 lb. | — |
| Resistance to water, chemicals, and cleaning solutions (Type IV) | 3.2.3.1; 3.2.3.3; 3.5 | 4.3.7; 4.3.8 | — | — | X | — | 1 | — | X |
| Yarns per inch: | | | | | | | | | |
| Warp, (ground and pile) | Table III | — | 5050 | — | — | X | 1 | — | X |
| Filling | Table III | — | 5050 | — | — | X | 1 | — | X |
| Loops per square inch | Table III | — | Visual | — | — | X | 1 | — | X |
| Tear strength | Table III | — | 5132 | — | — | X | 1 | .010 gr. | — |
| Weight, oz./sq. yd. | Table III | — | 5041[3] | — | — | X | 1 | 0.1 yd. | — |
| Resistance to boiling and cold water (Type V) | 3.2.4.2; 3.5 | 4.3.7 | — | — | X | — | 1 | — | X |
| Adhesion between sponge and mat | Table IV | — | 5970[2] | — | — | X | 1 | 0.1 lb. | — |
| Tensile strength, (sponge component) | Table IV | — | 5100[2] | — | — | X | 1 | 0.1 lb. | — |
| Tensile strength, (combined sponge and mat) | Table IV | — | 5100[2] | — | — | X | 1 | 0.1 lb. | — |
| Ash, (sponge component) | Table IV | 4.3.5 | — | — | — | X | 1 | 1.0 percent | — |
| pH range, (sponge component) | Table IV | 4.3.6 | — | — | — | X | 1 | 0.1 pH | — |
| Water absorption, (sponge component) | Table IV | 4.3.3 | — | — | — | X | 1 | 1.0 percent | — |
| Absorption after blotting, (sponge component) | Table IV | 4.3.4 | — | — | — | X | 1 | 1.0 percent | — |
| Resistance to water, chemicals, and cleaning solutions | 3.2.5.1; 3.2.5.2; 3.5 | 4.3.7; 4.3.8 | — | — | X | — | 1 | — | X |

[1] The contractor shall submit a certificate of compliance for this characteristic.

[2] Specimen shall consist of a piece of wet sponge or foam 1 inch in thickness by 2 inches wide and 2 inches long. To prepare the specimen, the sponge or foam shall be immersed in water at room temperature until it has expanded to its fullest volume and then removed from water and the excess squeezed out by hand and the specimen cut from the wet sponge or foam.

[3] At the option of the procuring officer, 1 square yard each of the uncoated and coated cloth shall be submitted for this test, or the contractor shall submit a certificate of compliance.



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4.3 Tests.

4.3.1 *Thickness measurement.* The thickness measurements for type II scouring pads shall be taken between the centers of two 2 by 4 by 0.045 inch (approximately) thick aluminum plates with a micrometer with no pressure. The average of at least three measurements shall be reported.

4.3.2 *Dimensions.* Types III and V scouring pads shall be immersed in water at room temperature until it is full of water and has expanded to the fullest volume. The pads shall be removed from the water, and the excess water squeezed out by hand. The dimensions of the wet pad shall be measured with a graduated scale to the nearest 1/4 inch in length and width, and to the nearest 1/8 inch in thickness.

4.3.3 *Water absorption.* (Types III and V.)

4.3.3.1 *Specimen.* The specimen shall be a piece of the sponge or foam 2 by 2 inches by the thickness of the sponge or foam.

4.3.3.2 *Apparatus.*

4.3.3.2.1 Equipment capable of maintaining the specimen at 23° ± 1.1°C. (73.5° ± 2°F.), and relative humidity of 50 ± 4 percent.

4.3.3.2.2 Analytical balance and weights.

4.3.3.2.3 Container of sufficient size to hold water for immersion of specimen.

4.3.3.2.4 Fine wire hook and support for suspending wet specimen.

4.3.3.3 *Procedure.* The specimen shall be washed free of water soluble materials by soaking it with tapwater and squeezing it as dry as possible, repeating the operation at least 5 times, using distilled water for the last operation. The washed specimen shall be conditioned for 24 hours at 23° ± 1.1°C. (73.5° ± 2°F.), and relative humidity of 50 ± 4 percent. The conditioned specimen shall be weighed to the nearest 10 milligrams and the weight recorded as $W_1$. The specimen shall then be immersed in water at room temperature, squeezed under water until the air is expelled, and then attached to a fine wire hook under water in such a manner that the original length by width area is held in a horizontal plane. The hook with specimen attached shall be removed from the water and suspended in air for 30 seconds. At the end of the 30 seconds, the specimen shall be weighed, the weight recorded at $W_2$, and the water absorbed calculated as follows:

$$\text{Water absorption, percent} = \frac{W_2 - W_1}{W_1} \times 100$$

Where: $W_1$ is the weight of the specimen before immersion, g.
$W_2$ is the weight of the specimen after immersion in water, g.

4.3.4 *Water absorption by weight blotting.* (Types III and V.)

4.3.4.1 *Specimen.* The specimen shall be a piece of the sponges or foam 2 by 2 inches by the thickness of the sponge or foam.

4.3.4.2 *Apparatus.*

4.3.4.2.1 Equipment capable of maintaining the specimen at 23° ± 1.1°C. (73.5° ± 2° F.), and relative humidity of 50 ± 4 percent.

4.3.4.2.2 Analytical balance and weights.

4.3.4.2.3 A square or rectangular tray for holding water, with dimensions of no less than twelve inches on each side, and capable of holding water at a depth of no less than one and one-half inches.

4.3.4.2.4 Fine wire hook and support for suspending wet specimen.

4.3.4.3 *Procedure.* The specimen shall be washed free of water soluble materials by soaking it with tapwater and squeezing it as dry as possible, repeating the operation at least five times, using distilled water for the last operation. The specimen shall be weighed immediately to the nearest 10 milligrams and the weight recorded as $W_1$. The tray shall be filled with cold tapwater to a height equal to one-fourth of the thickness of the specimen to be tested and placed on a horizontal level surface. The specimen shall then be placed gently in the tray with the thickness dimension vertical and the width-length dimensions in a plane parallel to the bottom of the tray. The specimen shall re-



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main partially immersed in the tray water at the specified depth of one-fourth the specimen thickness for a period of 15 seconds. The specimen shall then be removed from the water, attached to the fine wire hook, and suspended in the air for 30 seconds. At the end of 30 seconds, the specimen shall be weighed, the weight recorded as $W_2$, and the water absorbed calculated as follows:

$$\text{Water absorption, percent} = \frac{W_2 - W_1}{W_1} \times 100$$

Where: $W_1$ is the weight of the specimen before immersion, g.
$W_2$ is the weight of the specimen after immersion in water, g.

4.3.5 *Ash.* (Types III and V.)

4.3.5.1 *Specimen.* The specimen shall consist of approximately 2 g. of sponge or foam.

4.3.5.2 *Apparatus.*

4.3.5.2.1 Oven capable of maintaining the specimen at a temperature of 100° ± 2°C. (212° ± 3.6°F.).

4.3.5.2.2 Muffle furnace.

4.3.5.2.3 Analytical balance and weights.

4.3.5.2.4 Porcelain crucible.

4.3.5.2.5 Desiccator.

4.3.5.3 *Procedure.* The specimen shall be washed free of any water soluble material as described in 4.3.3.3, and dried to a constant weight in an oven at 100° ± 2°C. (212° ± 3.6°F.), cooled in a desiccator, weighed in a tared crucible, and the weight of the specimen recorded as $W_1$. The crucible with dried specimen shall be placed in a muffle furnace and gradually heated at low temperature so as to avoid flaming. After smoking has ceased, the temperature shall be raised rapidly until the specimen has burned free of carbonaceous material. The crucible and contents shall be removed from the furnace, cooled in a desiccator, weighed, the weight of the ash recorded as $W_2$, and the amount of ash in the specimen calculated as follows:

$$\text{Ash, percent} = \frac{W_2}{W_1} \times 100$$

4.3.6 Determination of pH (type V).

4.3.6.1 *Specimen.* The specimen shall consist of 2 g. of material taken from the cellulose sponge.

4.3.6.2 *Apparatus.*

4.3.6.2.1 pH meter.

4.3.6.2.2 Beaker, 300-ml.

4.3.6.2.3 Hotplate or gas burner.

4.3.6.3 *Reagent.* Distilled water.

4.3.6.4 *Procedure.* The specimen shall be immersed in 250 ml. of distilled water in a beaker. The solution shall be boiled for 10 minutes, cooled to room temperature, and the pH determined by means of a pH meter.

4.3.7 *Resistance to cold and boiling water.* Types II, III, and V test sample scouring pads shall be completely immersed in cold and boiling water at the temperature and time specified in table VII. The sample shall then be inspected for conformance with 3.5.

4.3.8 *Resistance to chemicals and alkalies.* Types II, III, and V test sample pads shall be completely immersed in the test solution at the temperature and time specified in table VII. The sample shall then be inspected for conformance with 3.5.

TABLE VII. *Immersion tests*

| Test solution | Temperature of test solution | Time (minutes) |
|---|---|---|
| Water, hot [1] | Boiling, 212°F. | 10 |
| Water, cold[2] | 25° ± 1°C. (77° ± 2°F.) | 30 |
| 5 percent NaOH[3] | 150 ± 5°F. | 5 |
| Mineral spirits solution (150° C-195°C.B.P. Range)[4] | 73 ± 5°F. | 10 |
| 15 percent trisodium phosphate solution[5] | 73 ± 5°F. | 10 |
| Mixture of one part general purpose cleaning compound and nine parts water.[6] | 73 ± 5°F. | 30 |
| Mixture of one part of abrasive soap cleaning compound and six parts water.[7] | 73 ± 5°F. | 30 |
| Mixture of one part heavy-duty synthetic detergent and three parts water.[8] | 73 ± 5°F. | 30 |



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TABLE VII. *Immersion tests (cont'd)*

| Test solution | Temperature of test solution | Time (minutes) |
|---|---|---|
| Mixture of one part solvent type detergent and three parts water[9] | 73 ± 5°F. | 30 |
| Mixture of one part general purpose detergent and three parts water.[10] | 73 ± 5°F. | 30 |

[1] The specimen shall be removed, allowed to cool, and the excess water squeezed out by hand and set aside to dry in air at room temperature. The specimen shall be thoroughly wet again with water and examined with that of the original specimen. (See 3.5.)

[2] The specimen shall be removed, the excess water squeezed out by hand, and set aside to dry in air. The specimen shall be thoroughly wet again with water and examined with that of the original specimen. (See 3.5.)

[3] Reagent shall conform to O-S-598.

[4] Reagent shall conform to TT-T-291.

[5] Reagent shall conform to O-S-642.

[6] Cleaning compound shall conform to P-C-435.

[7] Cleaning compound shall conform to P-C-442.

[8] Synthetic detergent shall conform to P-D-223.

[9] Solvent type detergent shall conform to P-D-225.

[10] General purpose detergent shall conform to P-D-220.

4.3.9 *Mineral grades.* The grading procedure of abrasive shall be in accordance with the U. S. Department of Commerce CS 217-59 (see 2.2).

## 5. PREPARATION FOR DELIVERY

5.1 **Packaging.** Packaging shall be level A, B, or C, as specified (see 6.2).

5.1.1 *Level A or B.* Pads of one type shall be unit packaged in the quantities shown in table VIII, in close-fitting boxes conforming to PPP-B-566, PPP-B-665, or to PPP-B-676, except each type V pad shall be placed in a plastic bag prior to packaging in a paperboard box. Each box shall be adequately sealed and secured to prevent accidental opening.

5.1.1.1 *Intermediate packaging.* Pads of one type shall be intermediate packaged in quantities shown in table VIII, in close-fitting boxes conforming to PPP-B-566, PPP-B-665, or to PPP-B-676. Each box shall be adequately sealed and secured to prevent accidental opening.

5.1.2 *Level C.* Pads shall be packaged in accordance with the supplier's commercial practice.

5.2 **Packing.** Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Unless otherwise specified (see 6.2), pads of one type, unit and intermediate packaged as specified in 5.1, shall be packed in the quantities shown in table VIII, in close-fitting boxes conforming to PPP-B-636, V3c. Each box shall be closed, sealed, waterproof sealed with tape, and reinforced with bands of filament tape in accordance with the appendix to the box specification.

5.2.2 *Level B.* Pads of one type shall be packed as specified in 5.2.1, except the box shall be class domestic and waterproof sealing and reinforcing with tape shall not be required.

5.2.3 *Level C.* The pads shall be packed to insure carrier acceptance and safe delivery to destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 **Standard pack (civil agencies).** For civil agency procurement, the standard pack shall be as specified in 5.1.1, 5.1.1.1, and 5.2.2.

5.4 **Marking.**

5.4.1 *Civil agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.4.2 *Military agencies.* In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.



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TABLE VIII

| | Unit package | Intermediate package | Shipping container |
|---|---|---|---|
| | Quantity per unit container | No. unit container per intermediate container | No. unit or intermediate cont. per shipping cont. |
| Type I | 1 | 12 | 6 |
| Type II | 10 | — | 6 |
| Type III | 12 (3 by 4-1/2 by 1/2") | 4 | 8 |
| | 24 (3-1/2 by 5 by 1-1/2") | 2 | 8 |
| Type IV | 1 | 8 | 20 |
| Type V | 5 | — | 8 |

## 6. NOTES

**6.1 Intended use.** Scouring pads are primarily intended for use in galleys, kitchens, and elsewhere in cleaning cooking utensils. Types II, III, and V scouring pads are also suitable for general cleaning and scouring purposes such as in building maintenance for washing walls, woodwork, and preparation of surfaces prior to painting.

**6.2 Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, and size required (see 1.2.1).

(*c*) Selection of applicable packaging and packing levels (see 5.1 and 5.2).

DOD coordination has been waived.

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.



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